C1 Experiment estimates:

Estimate T1: Since we are removing the whole sets average using the broadcast technique should drop off quickly to zero and then become constant

Estimate T2: Again all of the robots are able to communicate, but communication has to go through more channels, which means it will converge to zero quickly but not as quickly as T1

Estimate T3: First case where all robots are not able to hear each other, only their k nearest neighbors. As all the robots become closer, they will gain more and more neighbors and the value will again converge to zero semi quickly, slower than the first two cases.

Estimate T4 r min: This is the best case scenario, should del(k) should seem to decrease linearly when shown on a log scale graph

Estimate T4 r min +0.4 : This should show similar behavior to r min, with a slightly different intercept

C1 Experiment Results:

T1: The prediction for T1 is correct. All robots are able to see the average and should move to the average in one step with del(k) being zero with some error, which is exactly what happened. The gap between the two zones has little effect

T2: The star method also converged to the average very quickly but slightly slower than T1 as predicted. In the space that the star algorithm builds, each robot should be no more than 2 communications away from the next. For every step, the stars will move halfway towards the center and become more connected. With this specific outline, a specific robot's number of links does not matter, it is the spread of info across the hub that determines convergence speed. Again the gap between zones has little effect

T3: The logic for T3 was slightly incorrect. Each robot is only allowed to 2k neighbors, so information is spread slower with each time step which means more steps are needed to converge. Additionally, each of these steps is a local average so it takes more time for robots on opposite sides of the setup to communicate with each other on movements. Again, the gap between zones has little effect on the final result

T4 r min: This is the slowest convergence rate even though it has the same amount of edges as a ring. This is mainly because only a handful of rings cross the gap, which creates a bottleneck in information. I was correct that it would be slightly slower than using a larger radius value

T4 r min +0.4 : I was slightly wrong on this front because the larger radius creates more links for information to travel across different zones, so it does not exhibit the same slow convergence as the other T4 case

C.2 Spread and rendezvous time

Prediction for Protocol A: Since this protocol is a weighted sum, the weight of the main robot will make this convergence faster
Prediction for Protocol B: Since this is only a normal average, it will converge slower because the lead robots have less weight

Results: Our prediction was correct for protocol A and B, with the main reason for the difference being in the math. Since protocol A is based on a calculated degree weighted average of positions. The final difference ended up being pretty drastic, with protocol A meeting in 7 steps while protocol B takes 450 steps. A is a lot faster than b because each leaf only has one neighbor(being the initial robot) and is able to move to a midpoint between that robot and its initial position. This allows us to average all positions in terms of the lead robot with 1 step. Meanwhile protocol B covers a smaller distance per step which explains the difference in steps needed to reach convergence